BPM2_ESOFT QUIZ 3_S2026_GD

# of Questions: 30

Question #: 1

A 51-year-old woman is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation, intermittent blood in the stool, and an unintentional weight loss 6.8-kg (15-lb). Physical examination shows pallor of the oral mucosa and conjunctivae. Colonoscopy shows a mass in the descending colon that bleeds easily. Biopsy of the mass shows cells with multiple mitoses and loss of architecture. Which of the following best characterizes the affected gastrointestinal structure in this patient?

✓A. Haustra coli
B. Well-developed brush border
C. Plicae circularis
D. Intestinal villi
E. Paneth cells

Rationale: Teniae coli and haustra coli (the sacculations of the colon located between the teniae) are gross features of the colon.

Question #: 2

A 56-year-old man comes to the physician because of a 3-month history of nausea, weight loss and abnormal bowel movements. He appears emaciated. Physical examination shows abdominal distension and a palpable mass in the right upper quadrant. CT scan of the abdomen shows a tumor in the right hepatic flexure of the colon. Based on the location of the tumor, which of the following groups of lymph nodes will most likely receive metastasizing cells from the cancerous lesion?

A. Internal iliac
B. Inferior mesenteric
C. Celiac
✓D. Superior mesenteric
E. Superficial inguinal

Rationale: The right hepatic flexure is a derivative of the midgut. It will be drained by the midgut vein, which is the superior mesenteric vein. Therefore the lymphatic drainage will be to the superior mesenteric lymph nodes, since lymphatic drainage follows the venous drainage.

Question #: 3

A 75-year-old man comes to the physician for a follow-up examination. Two weeks ago, he underwent surgery for an abdominal aortic aneurysm. During surgery, the inferior mesenteric artery is excised. Which of the following arteries will most likely continue to supply blood to the distal colon through its anastomotic connections?

✓A. Middle colic
B. Ileal
C. Common hepatic
D. Jejunal
E. Splenic

Rationale: There is an anastomosis between the colic (left, right and middle) arteries via the marginal artery. Therefore if the inferior mesenteric artery is removed, the distal colon can still get it's blood supply.

Question #: 4

A 22-year-old African American man comes to the emergency department because of sudden-onset back pain, fatigue and dark urine. Three days ago, he was prescribed trimethoprim-sulfamethoxazole for a urinary tract infection. Physical examination shows scleral icterus and pale mucus membranes. Results of laboratory studies are shown in the table. The activity assay of a specific enzyme shows decreased enzymatic activity. The coenzyme formed from the deficient enzyme is most likely required for which of the following metabolic processes?

A. Pyruvate to oxaloacetate
B. Alpha-ketoglutarate to succinyl CoA
✓C. HMG CoA to mevalonate
D. Fatty acyl CoA to acetyl CoA

E. HMG CoA to acetoacetate

Rationale: The patient described has G6PD deficiency (pentose phosphate pathway). Note the hemolysis and Heinz bodies (RBC inclusions). The deficiency of pentose phosphate pathway results in low NADPH production. NADPH is required for fatty acid synthesis (for fatty acid synthase complex activity) and cholesterol synthesis (HMG CoA reductase and conversion of lanosterol to cholesterol). The pathway that is affected in this patient would be cholesterol synthesis. HMG CoA reductase of cholesterol synthesis requires NADPH (as reducing equivalent).

Conversion of pyruvate to oxaloacetate (Pyruvate carboxylase) requires biotin
Alpha-ketoglutarate to succinyl CoA (Alpha-keto glutarate dehydrogenase) requires NAD+, TPP, CoA, lipoic acid and FAD.
Fatty acyl CoA to acetyl CoA (beta-oxidation) uses FAD and NAD.
HMG CoA to acetoacetate (ketone body synthesis does NOT require NADPH.

Hemoglobin	8.5 g/dL (normal: 13.5 – 17.5)
Reticulocytes	7% (normal: 0.5% - 1.5%)
Urinalysis **Blood** **Red blood cells**	 Moderate 0/hpf
Liver function studies **Total bilirubin** **Direct bilirubin**	 3.5 mg/dL (0.1 – 1.0 mg/dL) 0.8 mg/dL (0.0 – 0.3 mg/dL)
Lactate dehydrogenase	700 U/L (45 – 200 U/L)
Peripheral smear	Bite cells and red blood cell inclusions on crystal violet staining

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Question #: 5

A 45-year-old man comes to the physician for a health maintenance examination. He says he had a burger with fries and a soft drink for lunch 10 minutes ago and is extremely drowsy. Physical examination shows no abnormalities. The physician explains that his metabolism is increased and there is increased activity of specific biochemical pathways. Which of the following enzymes and its corresponding substrate is most likely utilized during his current state?

A. I
B. II
C. III
D. IV
E. V
✓F. VI
G. VII
H. VIII

Rationale: The person is in the well-fed state and insulin glucagon ratio is elevated.
I - Pyruvate carboxylase (enzyme of gluconeogenesis) is less active; Converts pyruvate to oxaloacetate and is the first enzyme of gluconeogenesis
II - Hormone sensitive lipase (Enzyme required for adipose tissue lipolysis) - active when the insulin : glucagon ratio is low (fasting); Breaks down TAG to fatty acids for beta-oxidation for use by liver and muscle
III - Glycogen phosphorylase (Enzyme of glycogenolysis); Breaks down glycogen to glucose 1-phosphate which is later converted to glucose in liver; Active in the fasting state
IV - Fatty acid synthase complex (multienzyme complex) - required for fatty acid synthesis; Active in the fed state; Substrates used are acetyl CoA and Malonyl CoA; Palmitate is the product of this enzyme
V - Succinyl CoA acetoacetate CoA transferase (Thiophorase) - Enzyme in peripheral tissues (brain and muscle) for activation and utilization of ketone bodies; Active in the fasting state
VI - Citrate lyase - Correct answer - In the well-fed state, mitochondrial citrate enters the cytosol and forms acetyl CoA in the cytosol for utilization in fatty acid and cholesterol synthesis; both pathways are active in the fed state; Acetyl CoA carboxylase (fatty acid synthesis) is activated by citrate and is active in the dephosphorylated state (insulin: glucagon ratio is high); HMG CoA reductase is also active in the dephosphorylated state (High insulin: glucagon ratio. Acetyl CoA carboxylase and HMG CoA reductase are inhibited in the phosphorylated state (low insulin: glucagon ratio --> Activates AMP-kinase which phosphorylates both enzymes and they are less active)
VII - Glycogen synthase is active in the liver and muscle in the fed state; but the substrate for glycogen synthase is UDP-glucose (active glucose)
VIII - Mitochondrial HMG CoA synthase (regulated enzyme of ketone body synthesis; found in liver) - active in the fasting state - Forms HMG CoA by condensation of acetoacetyl CoA and acetyl CoA

	Enzyme activity	Substrate
I	Pyruvate carboxylase	Pyruvate
II	Hormone sensitive lipase	Adipose TAGs
III	Glycogen phosphorylase	Glycogen
IV	Fatty acid synthase complex	Palmitate
V	Succinyl CoA: acetoacetate CoA transferase (Thiophorase)	Succinyl CoA
VI	Citrate Lyase	Citrate
VII	Glycogen synthase	Lactate
VIII	Mitochondrial HMG CoA Synthase	Acetyl CoA

Question #: 6

A 17-year-old girl is brought to the physician by her parents because of a 1-day history of right flank pain and red-colored urine. Imaging studies show an opaque mass in her right kidney. Urinalysis shows the presence of RBCs and hexagonal crystals. Deficiency of which of the following transporters is most likely responsible for this patient's condition?

✓A. Basic amino acid transporter
B. Branched chain amino acid transporter
C. Aromatic amino acid transporter
D. Acidic amino acid transporter
E. Neutral amino acid transporter

Rationale: This patient most likely has cystinuria. The most common cause of cystinuria is due to a deficiency in the basic amino acid transporter. Mutations of the basic amino acid transporter gene associated with cystinuria. The basic amino acid transporter transports (COAL OR COLA) lysine; arginine; histidine; and cysteine amino acids. Improper reabsorption of cysteine causes accumulation of cysteine in the kidneys. The 2 cysteine residues oxidizes to form cystine; which is not water soluble therefore; precipitates as cystine crystals during excretion.
Hartnup disease is due to deficiency of the neutral amino acid transporter. Presents as growth retardation (as TRP is an essential amino acid) and niacin deficiency (Pellagra; NAD is synthesized from Trp)

Question #: 7

A 10-day-old boy is brought to the emergency department by his parents because of a 6-hour history of lethargy, poor feeding and progressive unresponsiveness.

He was born by vaginal delivery at-term and there were no abnormalities at birth. Laboratory studies show markedly elevated blood ammonia levels and respiratory alkalosis. Genetic tests show an inborn error of metabolism. Which of the following enzyme deficiency disorders is most likely in this patient?

A. Glucose 6-phosphatase
B. Medium chain acylCoA dehydrogenase
C. Carnitine palmitoyl transferase-I (CPT-I)
✓D. Carbamyl phosphate synthetase-I (CPS-I)
E. Bilirubin UDP-glucuronyl tranferase

Rationale: The infant in the vignette has a urea cycle disorder. Note the elevated blood ammonia levels. There is no elevation in the other urea cycle intermediates and hence it is most likely an inherited deficiency of carbamyl phosphate synthetase-1 (CPS-1).
CPT-I deficiency manifests as hypoglycemia and hypoketosis. Systemic fatty acid oxidation defect affecting beta-oxidation in the liver.
Bilirubin -UDP glucuronyl transferase deficiency can present with jaundice and elevated levels of unconjugated bilirubin. Physiological jaundice (low activity of the enzyme) usually resolves spontaneously. Gilbert syndrome is characterized by partial deficiency of the enzyme. Crigler-Najjar syndrome is characterized by more severe deficiency of the enzyme.
Glucose 6-phosphatase deficiency presents with hypoglycemia and lactic acidosis.
MCAD deficiency presents as hypoglycemia and hypoketosis.

Question #: 8

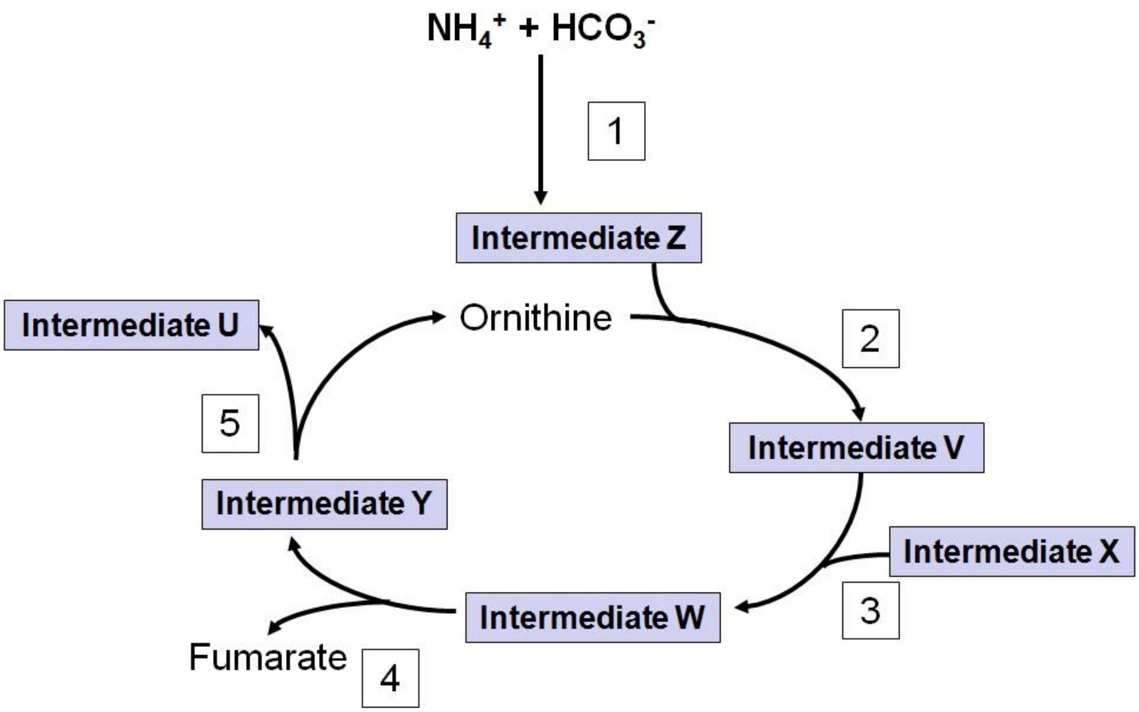
A 4-month-old boy is brought to the emergency department by his parents because of a 1-day history of seizures. An inborn error of metabolism involving the pathway that is deficient in this patient is shown. Laboratory studies show high blood levels of argininosuccinate. Which of the following best represents the intermediate accumulating in this patient?

A. Intermediate U
B. Intermediate Z
C. Intermediate U
✓D. Intermediate W
E. Intermediate V

Rationale: The patient has argininosuccinate lyase deficiency (Enzyme 4 in the pathway). As a result of the enzyme deficiency, argininosuccinate levels are elevated.
The diagram shows the urea cycle.
Intermediate Z: Carbamoyl Phosphate which is formed by Carbamoyl phosphate synthetase-I

(CPS-I) which is labeled as Enzyme 1 in the pathway.
Intermediate V: Citrulline formed by Ornithine transcarbamylase (Enzyme 2 in the pathway)
Intermediate X: Aspartate (which donates a Nitrogen atom for Urea formation. Enzyme 3 is Argininosuccinate synthetase.
Intermediate W: Argininosuccinate which is elevated in this patient.
Enzyme 4 is Argininosuccinate Lyase which cleaves argininosuccinate into Fumarate and Arginine (Labeled as Intermediate Y)
Intermediate Y: Arginine which is cleaved by Enzyme 5 (Arginase) into Urea (Intermediate U) and Ornithine.
Intermediate U: Urea

Enzymes 1 and 2 are mitochondrial enzymes; Enzymes 3,4 and 5 are cytosolic enzymes.



Question #: 9

A 3-month-old boy is brought to the physician because of a 5-day history of persistent crying and severe skin blisters. Physical examination shows blisters all over his body in various stages of healing. Urinalysis shows red color urine and elevated levels of porphyrins type I. Which of the following is the most likely diagnosis?

A. Lead poisoning
B. Pyridoxine deficiency
C. Acute intermittent porphyria
✓D. Congenital erythropoietic porphyria
E. Porphyria cutanea tarda
F. Gilbert syndrome

Rationale: This is a patient with congenital erythropoietic porphyria; Note the early age of presentation and skin manifestations. They also have photophobia and increased hair growth (hypertrichosis). Uroporphyrinogen-III synthase isozyme in bone marrow is deficient in Congenital Erythropoietic Porphyria which shows accumulation of red uroporphyrin-I and coproporphyrin-I in tissues, blood and urine.

Acute intermittent porphyria presents as severe abdominal pain and neuropsychiatric manifestations. Deficiency of porphobilinogen deaminase (hydroxymethylbilane synthase. Patients do NOT have cutaneous manifestations. Excretion of delta-ALA and porphobilinogen (purple color) in urine.

Porphyria cutanea tarda is characterized by deficiency of uroporphyrinogen decarboxylase. Uroporphyrin-III accumulates in blood and urine. Characterized by photosensitivity and onset typically in adults. Ethanol use and sun exposure increase the risk of symptomatic presentation.

Pyridoxine deficiency results in microcytic anemia due to reduced activity of delta-ALA synthase and reduced heme synthesis.

Lead poisoning inhibits delta-ALA dehydratase and ferrochelatase. Characterized by anemia and reduced heme synthesis.

Gilbert syndrome is characterized by episodes of jaundice and elevations in indirect (unconjugated) bilirubin usually following an illness or stress. Characterized by about 50% activity of UDP-glucuronyl transferase.

Question #: 10

A 4-week-old boy is brought to the physician by his mother because of a 3-day history of vomiting and failure to thrive. He is exclusively breast-fed. The mother says that vomiting usually occurs within an hour of breast feeding. Physical examination shows jaundice and an enlarged liver with a palpable lower border of the liver. He has a developmental delay. The blood glucose level is 50 mg/dL. Urinalysis shows the presence of a reducing sugar, that is not glucose. Which of the following enzyme deficiency disorders is the most likely diagnosis in this infant?

A. β-galactosidase
B. Fructokinase
C. Galactokinase
✓D. Galactose 1-phosphate uridyltransferase
E. Aldolase B
F. Branched chain keto acid dehydrogenase

Rationale: The vignette describes a child with classical galactosemia due to GALT deficiency. Note the hypoglycemia on breast feeding, involvement of the liver (jaundice and hepatomegaly) and neurological developmental delay. Presence of a reducing sugar in the urine is an indication that it is most likely galactose in urine.

Question #: 11

A 25-year-old woman is brought to the physician by her friend because of a 20-day history of starving herself. Physical examination shows muscle atrophy and poor grip strength. She is observed to have muscle protein catabolism. Which of the following amino acids most likely forms only acetyl CoA or ketone bodies on catabolism?

A. Alanine
B. Tyrosine
C. Glutamine
D. Aspartate
✓E. Leucine
F. Glycine

Rationale: The amino acids that are only ketogenic are leucine and lysine. All other amino acids are only glucogenic, or both ketogenic and glucogenic.

Question #: 12

A 2-month-old girl is brought to the physician by her parents because of a 2-day history of intermittent seizures. The mother says the urine has a mousy (musty) odor. Physical examination shows developmental delay. She is prescribed a special diet lacking an amino acid. Her mother is also advised to avoid aspartame in the infant's diet. Which of the following inherited disorders of metabolism is the most likely diagnosis?

A. Alkaptonuria
✓B. Phenylketonuria
C. Homocystinuria
D. Hyperammonemia type I
E. Maple syrup urine disease
F. Medium chain acyl CoA dehydrogenase deficiency
G. Galactosemia
H. Lactose intolerance

Rationale: The vignette describes a child with phenylketonuria. The musty odor of the urine is due to the presence of phenylacetate and phenyllactate in the urine.

Question #: 13

A 25-year-old man has been starving for the past 20 days and shows significant denovo synthesis from amino acids. Which of the following amino acids would directly form oxaloacetate as a precursor of glucose by gluconeogenesis?

A. Alanine
B. Tyrosine
C. Glutamine
✓D. Aspartate
E. Leucine
F. Glutamate

Rationale: Oxaloacetate is formed from the aminoacid, aspartate. Alanine forms pyruvate. Glutamate forms alpha-keto glutarate. Leucine is a ketogenic amino acid.

Question #: 14

Metabolic changes in a 22-year-old fasting woman is being studied. She only sips approximately 100 ml of apple juice throughout day 1 of fasting. Which of the following is most likely observed on the second day of the fast?

A. The rate of β-oxidation is decreased in the liver
B. The activity of lipoprotein lipase is increased
C. Hepatic malonyl CoA levels are increased

✓D. CPT-1 activity is increased in the liver
E. Glycerol is utilized for energy generation in the adipose tissue
F. Hormone sensitive lipase is in the dephosphorylated form
G. Mitochondrial HMG-CoA synthase activity is low
H. Hepatic fructose 2,6 bisphosphate levels are elevated

Rationale: The person in the vignette is in starvation with very low caloric intake (50 Cals). During starvation, adipocyte lipolysis is increased, beta oxidation in tissues is activated (due to increased activity of CPT-1), key enzymes are in the phosphorylated form (under the influence of glucagon), hepatic ketogenesis is active (mitochondrial HMG CoA synthase), liver glycolysis is less active and liver gluconeogenesis is more active (low levels of fructose 2,6-bisphosphate). Lipoprotein lipase is more active in the fed state.

Question #: 15

A 28-year-old woman comes to the physician because of a 3-month history of tiredness and easy fatigability. Physical examination shows conjunctival pallor. Laboratory studies show a hemoglobin of 9.2 g/dL (reference range 11.4-16.0 g/dL). A peripheral blood film shows macrocytic, megaloblastic anemia. Further tests show that almost all the folate is in the methyl-tetrahydrofolate form. Which of the following additional laboratory findings is most likely in this patient?

A. Increased serum protoporphyrin levels
✓B. Increased serum methylmalonate levels
C. Increased urinary copper excretion
D. Low serum ferritin levels
E. Decreased red cell transketolase activity

Rationale: The vignette describes a patient with vitamin B12 deficiency. Note the occurrence of folate trap as methyltetrahydrofolate. Many patients may also have accompanying neurological features. An important laboratory finding in B12 deficiency is elevated levels of serum methylmalonate.

Question #: 16

A 43-year-old man comes to the physician because of a 6-month history of brownish-black discoloration of his ears and nose. He says it has slowly progressed in the last year and has become quite apparent in the past few weeks. He has a history of dark urine since birth. Accumulation of which of the following compounds is most likely responsible for this clinical finding?

A. Melanin
✓B. Homogentisic acid
C. Homocysteine
D. Coproporphyrin-III
E. Porphobilinogen
F. Cystine

Rationale: The vignette describes a patient with alkaptonuria (homogentisate oxidase deficiency). The patient has ochronosis and joint pain due to accumulation of homogentisic acid in cartilage.

Question #: 17

A 10-day-old boy is brought to the emergency department by his parents because of a 6-hour history of lethargy, poor feeding and progressive unresponsiveness. He was born by vaginal delivery at-term and there had no abnormalities at birth. Laboratory studies show markedly elevated blood ammonia levels and respiratory alkalosis. Genetic tests show an inborn error of metabolism. He is prescribed a drug to facilitate removal of excess nitrogen from his body. Which of the following drugs was most likely administered to this patient?

A. N-acetyl cysteine
B. Acetoacetic acid
C. Glutamine
✓D. Benzoic acid
E. Aspartate
F. Carnitine
G. Hemin
H. L-DOPA

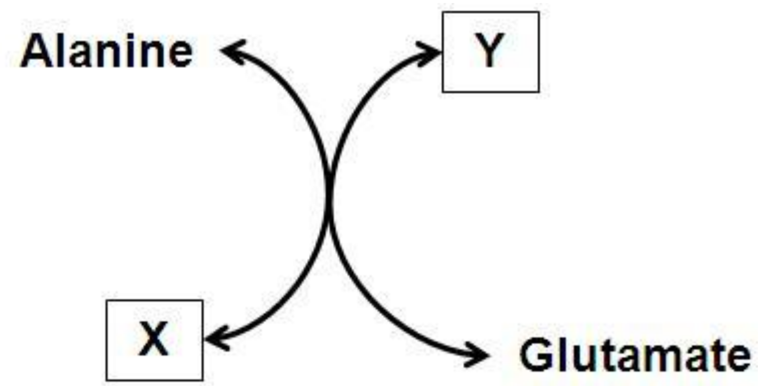
Rationale: The infant in the vignette has a urea cycle disorder. Administration of benzoic acid conjugates with glycine to form hippuric acid which is excreted in urine. Phenylacetate may also be given which combines with glutamine and is excreted as phenylacetylglutamine. Benzoic acid and phenylacetate are ammonia scavenging drugs and are useful as alternative pathways for nitrogen disposal.

Question #: 18

An investigator studying amino acid metabolism isolates an enzyme from cultured hepatocytes. The reaction catalyzed by the enzyme is shown. Which of the following best describes this reaction?

A. X represents ammonia and Y represents glutamine
✓B. X and Y are ketoacids of alanine and glutamate, respectively
C. The coenzyme required for this reaction is NAD^{+}
D. Patients with thiamine deficiency have low activity of this enzyme
E. X is alpha-ketoglutarate and Y is oxaloacetate

Rationale: The reaction is catalyzed by alanine aminotransferase (ALT). X is pyruvate and Y is alpha-ketoglutarate. Vitamin B6 (PLP) is required as a coenzyme for this reaction. Note that free ammonia is NOT liberated in this reaction.



Question #: 19

An investigator is studying the regulation of an enzyme-catalyzed reaction shown. The activity of the enzyme is observed under different metabolic conditions. Which of the following most likely inhibits the activity of this enzyme?
HMG CoA → Mevalonate

✓A. Phosphorylation
B. Acetyl-CoA
C. Cholesterol ester
D. Citrate
E. Dephosphorylation
F. NADPH
G. Insulin

Rationale: The enzyme catalyzing the reaction is HMG CoA reductase which is the key regulated enzyme of cholesterol biosynthesis. Phosphorylation of the enzyme results in reduced activity of the enzyme. NADPH is a required coenzyme for the reaction and the activity of the enzyme is increased by insulin.

Question #: 20

A 23-year-old man comes to the physician because of a 2-day history of worsening bloating, diarrhea, dehydration and abdominal pain. He says the symptoms occur a few hours after eating meals with cereal, milk and sugar. Four days earlier, he had an episode of seafood poisoning with severe diarrhea, dehydration and abdominal pain. Which of the following is the best explanation for the worsening of the symptoms in this patient?

A. Increased activity of enteropeptidase
B. Decreased secretion of pancreatic amylase
✓C. Decreased activity of intestinal disaccharidases
D. Decreased secretion of bile salts by the gall bladder
E. Increased secretion of gastrin by the stomach

Rationale: The vignette describes a patient with secondary lactase deficiency. Lactase is an enzyme that hydrolyzes the beta1-4 links in the disaccharide lactose. The disaccharidases are present on the brush border of the intestinal villi.

Question #: 21

A 50-year-old woman comes to the physician because of a 3-month history of weight loss. Physical examination shows conjunctival icterus and tenderness in the right epigastric region. Laboratory studies show elevated levels of serum total and direct bilirubin. Indirect bilirubin levels are normal. Serum ALP levels are increased; but serum amylase and lipase levels are normal. Ultrasound examination of the abdomen shows gall stones obstructing the common bile duct. Which of the following steps is most likely directly affected as a consequence?

A. Activation of pepsinogen to pepsin
B. Digestion of dietary starch
✓C. Absorption of dietary triacylglycerol
D. Formation of chylomicrons
E. Activation of trypsinogen to trypsin
F. Absorption of dietary vitamin B12

G. Digestion of disaccharides

Rationale: The patient in the vignette shows obstructive jaundice and reduced bile salt entry into the intestine. As a result, there is impaired digestion of dietary lipids (as emulsification is defective) and impaired absorption of the dietary lipids (as bile salts are required for the formation of micelles)

Question #: 22

A 16-year-old boy is brought to the physician by his parents because of a 2-day history of yellowing of the eyes and skin. Two days earlier, he was prescribed a sulfa drug for a respiratory infection. A provisional diagnosis of inherited glucose 6-phosphate dehydrogenase deficiency is made. Which of the following sets of laboratory results will most likely be found in this patient?

A. I
B. II
✓C. III
D. IV
E. V

Rationale: The patient has hemolytic anemia due to G6PD deficiency. Hemolysis is triggered by administration of sulfa drugs (oxidative drugs). Hemolytic anemia is characterized by elevated levels of unconjugated bilirubin, absence of bilirubin in urine (acholuric jaundice) and increased urobilinogen in urine (due to increased conjugation of bilirubin in the liver)

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Unconjugated	Absent	Decreased
II	Unconjugated	Present	Increased
III	Unconjugated	Absent	Increased
IV	Conjugated	Absent	Normal
V	Conjugated	Present	Decreased

Question #: 23

A 25-year-old man comes to the physician because of a 1-month history of intermittent abdominal cramps and bloating. History shows that the symptoms typically appear following meals rich in carbohydrates. Physical exam shows no abnormalities. Genetic testing shows a mutation in a gene which encodes for an enzyme that breaks down disaccharides to monosaccharides for absorption. Which of the following is most likely the typical localization of this enzyme?

A. Digestive organelles of hepatocytes
B. Cytoplasm of enteroendocrine cells
✓C. Microvilli of enterocytes
D. Glands of the intestinal submucosa
E. Lumen of the stomach

Rationale: .

The plasma membrane of the microvilli of the enterocyte plays a role in digestion as well as absorption. Digestive enzymes are anchored in the plasma membrane, and their functional groups extend outward to become part of the glycocalyx. This arrangement brings the end products of digestion close to their site of absorption. Included among the enzymes are peptidases and disaccharidases. These enzymes contribute to the digestive process by completing the breakdown of most sugars and proteins to monosaccharides and amino acids, which are then absorbed.

Question #: 24

A 56-year-old woman is brought to the emergency department because of a 6-week history of fatigue, dizziness, and intermittent chest pain. History shows that she has Graves' disease. Her blood pressure is 90/60 mmHg and pulse is 120/min. Physical examination shows pale conjunctivae and oral mucosa and macroglossia (enlarged tongue). Laboratory studies of her blood show a hemoglobin level of 11 g/dL (normal range: 12-16 g/dL) and a vitamin B12 level of 165ng/mL (normal range 200-900 ng/mL). Which of the following is most likely to be an additional finding and a potential cause of the abnormalities seen in this patient?

A. Decreased lysosomal functions

B. Increased secretion of chymotrypsin
✓C. Antibodies against intrinsic factor
D. Increased secretion of mucus
E. Hyperplasia of chief cells

Rationale: This is a person with pernicious anemia. The fact that she has an existing autoimmune condition places her at risk for the development of another. Pernicious anemia has many causes including gastric bypass and dietary deficiencies, but it is also known to be cause by autoimmune antibodies against either the parietal cells (which produce intrinsic factor) or directly against intrinsic factor. Pernicious anemia usually has a deficiency of vitamin B12 because these patient cannot bind B12 in the stomach to carry it to the small intestine for absorption. Vitamin B12 is needed for adequate red blood cel synthesis

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Question #: 25

A 27-year-old woman comes to the physician because of a 1-week history of loss of appetite, nausea, weakness, and dark urine. Her temperature is 38ºC (101ºF), pulse is 100/min and respirations are 22/min. Physical examination shows right upper quadrant abdominal tenderness. Laboratory studies of the blood show antibodies against hepatitis B virus. A diagnosis of acute hepatitis B infection is made. During this disease there is initial activation of the cells shown on the attached micrograph. Which of the following is the most likely function of these cells?

A. Production of bilirubin
✓B. Phagocytosis of red blood cells
C. Storage of vitamin A as retinyl esters
D. Degradation of exogenous toxins and drugs
E. Synthesis of steroid hormones

Rationale: The image is a TEM of hepatic sinusoids with the arrow pointing to a stellate sinusoidal macrophage (Kupffer Cell), a member of the mononuclear phagocytic system. The presence of red cell fragments and iron in the form of ferritin in the Kupffer cell cytoplasm suggests that they are involved in the final breakdown of damaged or senile red blood cells that reach the liver from the spleen

Question #: 26

A 64-year-old man comes to the physician because of a 3-week history of epigastric pain, abdominal discomfort, and weight loss. Physical examination shows no abnormalities. Laboratory studies of gastric juice show an abnormally high gastric acid output. The serum gastrin level is not elevated. Gastric mucosal studies show increased production of histamine in the stomach. Which of the following cells is the source of this substance?

✓A. ECL cells
B. S cells
C. Chief cells
D. D cells

E. G cells

Rationale: ECL cells release histamine that stimulates parietal cells to secrete acid Option A). S cells secrete secretin that stimulates pancreatic HCO3-. Chief cells release digestive enzymes (pepsin & lipase). D cells release somatostatin that inhibits gastric acid release. G cells secrete gastrin that stimulates acid secretion; but the gastrin level is not elevated in this patient.

Question #: 27

A 74-year-old woman comes to the physician for a follow-up examination. Six months ago, she was diagnosed with an early stage of gastric carcinoma in the upper stomach and a partial gastrectomy involving the fundal region of the stomach was performed. Physical examination shows no abnormalities. Which of the following functions of the stomach is most likely affected in this patient?

A. Acid secretion
✓B. Receptive relaxation
C. Basic electrical rhythm
D. Migrating motor complex
E. Mixing and grinding

Rationale: Food arriving at the stomach induces receptive relaxation in the fundus (upper part) of the stomach. Since this area has been surgically removed, receptive relaxation would be most affected.

Peristalsis, the BER, mixing, and acid secretion all start from the body (middle part) of the stomach, so they would not be significantly affected. The MMC starts from the lower part of the stomach into the small intestines and would be unaffected by surgical removal of upper part of the stomach.

Question #: 28

A 19-year-old man participates in a study related to the motility of the alimentary tract. He is given a single dose of the hormone motilin, which triggers peristaltic waves that start from the duodenum and move towards the caecum. The options of contraction and relaxation of smooth muscles in oral and caudal direction to the food are shown in the attached table. Which of the following options of smooth muscle contraction and relaxation is most likely during intestinal peristalsis?

A. A
B. B
✓C. C
D. D

Rationale: Motilin stimulates the receptor agonists to stimulate intestinal peristalsis. In peristaltic contraction such as MMCs, a section of small intestine contracts 'behind/orad' to the chyme. Simultaneously, a section of small intestine relaxes 'ahead/caudad' of the chyme. This propels any residual intestinal contents along the gastrointestinal tract.

	Smooth Muscle Oral of the Food	Smooth Muscle Caudal of the Food
A	Contracted	Contracted
B	Relaxed	Contracted
C	Contracted	Relaxed
D	Relaxed	Relaxed

Question #: 29

A 22-year-old man volunteers to be a subject in a study on luminal fluid pH in the stomach. A pH-sensitive microelectrode is slowly advanced from the lumen of the stomach deep into the gastric mucosal surface. Results of change in pH as the microelectrode moves from the lumen deep inside is shown in the attached figure. It has been observed that the pH increases as the electrode approaches the gastric surface. Which of the following secretory changes best explains the observed increase in pH of the fluid?

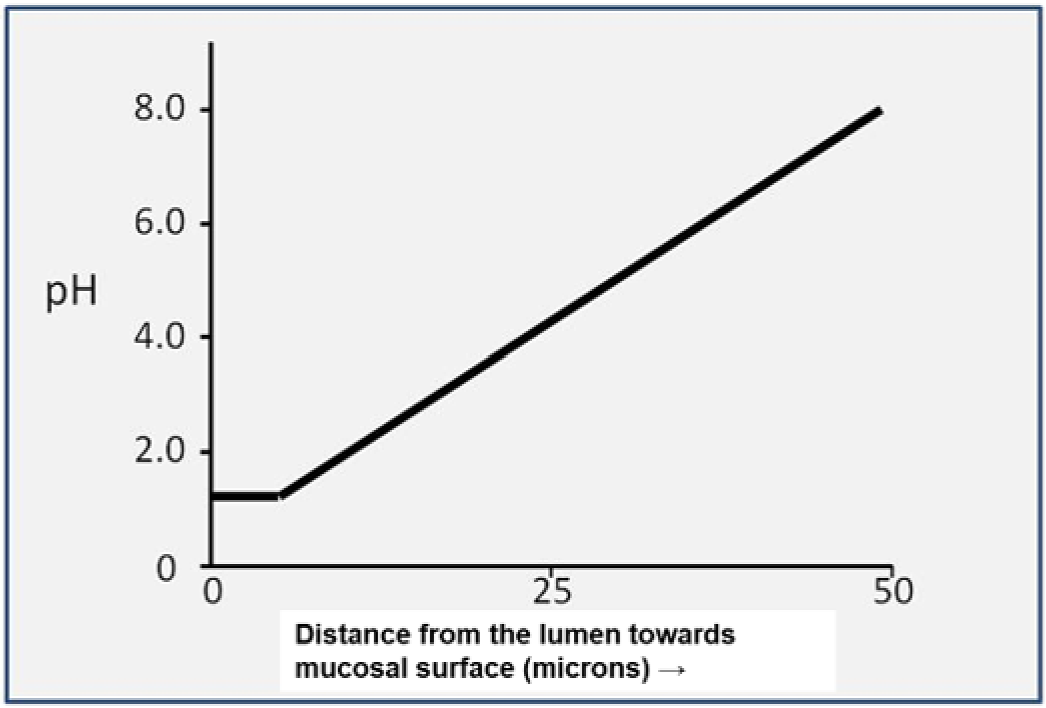
A. Increased release of acetylcholine from the vagus nerve
B. Release of gastrin from the G cells

✓C. Secretion of bicarbonate ions by the mucus cells in the surface epithelium

D. Secretion of H^+ by mucus cells in the surface epithelium

E. Release of secretin from duodenal S-cells

Rationale: The low pH (high acidity) in the lumen of the stomach is due to secretion of H+, which in turn is stimulated by secretin, gastrin and ACh from the vagus. However, the presence of luminal acidity does not explain the gradient towards a more neutral pH (increasing toward 7.4, i.e. less acidic) as we approach the mucosal surface, as shown in the figure. This gradient is due to bicarbonate release from mucus cells (correct answer), and it is supported by a mucus lining over the epithelial surface, which traps the bicarbonate, preventing it from leaking into the lumen.



Question #: 30

A 30-year-old woman comes to the physician because of a 6-month history of progressive difficulty swallowing and a 4-kg (8.8-lb) weight loss. Physical examination shows no abnormalities. A manometric study is conducted to examine pressure generation along the length of her esophagus. Manometry shows that esophageal contractions in response to a swallow are poorly synchronized and the pressure in the lower esophageal sphincter remains elevated at rest as well as during swallowing. Which of the following is the most likely diagnosis?

A. Gastroesophageal reflux disease

✓B. Achalasia
C. Hiatal Hernia
D. Scleroderma
E. Barrett's esophagus

Rationale: The lack of a coherent esophageal motility pattern, combined with a failure of the lower esophageal sphincter to relax fully to allow swallowing to complete, are the two key detailed features of Achalasia (besides the more general observation of difficulties with swallowing). Accordingly, Achalasia is the correct answer.